

Literature Resolution: The Discrete John Condition Is Unnecessary for Thin Complements

Run fa_banach_001 — lane 1

13 August 2026

Status: literature already answered (fully). The question in arXiv:1211.6933 asks whether a discrete John/accessibility condition is really needed to deduce a Hardy inequality and quasiadditivity of variational capacity when the complement has Aikawa codimension greater than p . Theorem 1.1 and Corollary 1.4 of arXiv:1402.6134 remove that condition in the same metric-space setting.

1 The source question

J. Lehrbäck and N. Shanmugalingam, *Quasiadditivity of variational capacity*, arXiv:1211.6933, work in a complete doubling metric measure space X supporting a p -Poincaré inequality, with $1 < p < \infty$. For an open set $\Omega \subset X$, their Proposition 4.1 proves that

$$\text{codim}_A(X \setminus \Omega) > p$$

implies both a p -Hardy inequality and quasiadditivity of the relative variational p -capacity, provided that Ω satisfies an additional discrete John condition for its Whitney cover.

Immediately after Proposition 4.1, on PDF page 13, they ask:

Source question. *Is the discrete John condition really necessary in Proposition 4.1?*

They knew no counterexample and observed that the condition was already unnecessary in Euclidean space.

2 The later resolution

J. Lehrbäck, *Hardy inequalities and Assouad dimensions*, arXiv:1402.6134, uses the notation $\underline{\text{codim}}_A$. Remark 2.1 identifies this lower Assouad codimension with the integral notion called the *Aikawa codimension* in the source paper.

Specializing Theorem 1.1 of the later paper to $\beta = 0$ gives:

Lehrbäck's theorem and corollary. *Let $1 < p < \infty$. Assume that X is an unbounded complete doubling metric measure space supporting a p -Poincaré inequality. If $\Omega \subset X$ is open and*

$$\underline{\text{codim}}_A(\Omega^c) > p,$$

then Ω admits a p -Hardy inequality. Consequently, the variational p -capacity $\text{cap}_p(\cdot, \Omega)$ is quasiadditive with respect to Whitney covers $\mathcal{W}_c(\Omega)$ for sufficiently small $c > 0$.

The consequence is stated verbatim as Corollary 1.4 of the later paper. The paragraph immediately preceding that corollary explicitly explains that Theorem 1.1 makes the earlier “additional accessibility condition” unnecessary.

3 Why this answers the exact question

There is no gap in terminology or hypotheses:

- the source’s codim_A is precisely the later paper’s lower Assouad codimension;
- setting $\beta = 0$ in the later theorem gives the same unweighted p -Hardy inequality;
- the ambient assumptions are the same complete, unbounded, doubling, p -Poincaré setting; and
- Corollary 1.4 states the desired Whitney-cover quasiadditivity directly, with no discrete John hypothesis.

Thus the answer is: *no, the discrete John condition is not necessary*. The codimension assumption alone suffices under the standing metric-space hypotheses. No new mathematical result is claimed here.

References

- [1] J. Lehrbäck and N. Shanmugalingam, *Quasiadditivity of variational capacity*, Potential Analysis 40 (2014), 247–265; arXiv:1211.6933.
- [2] J. Lehrbäck, *Hardy inequalities and Assouad dimensions*, Journal d’Analyse Mathématique 131 (2017), 367–398; arXiv:1402.6134; doi:10.1007/s11854-017-0013-8.